LANCE

An autonomous and verifiable auditor for IoT/IIoT networks

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The objective

Under which conditions does an agentic approach prevail over a scripted one, rather than which model prevails?

- An auditor that explores, identifies, verifies and reports.
- Identical tools and budget for both approaches, within the control harness.
- Evidence recomputed from the log, not asserted.

Model training is not a current priority.

The platform in brief

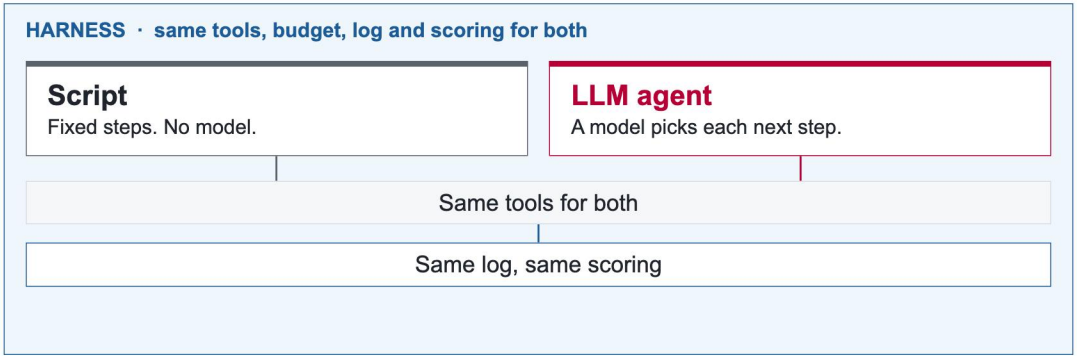
- **Runs the audit in six phases**
- **Four execution modes**
- **Every run kept as evidence**



The platform runs the audit; the benchmark judges it.

Captures of the running instance, 22 October 2026.

Script, LLM agent and harness



Only the decision-maker changes.

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How a run is judged against the goal

1 Goal
Find the weaknesses of a network, and only those

2 Laboratory
A scenario with planted weaknesses, listed in a ground-truth file

3 Platform
Audits the scenario: six phases, one log per run

4 Evidence
Observations and proofs saved as files for each run

5 Benchmark
Compares findings with the truth: precision, recall, F1

A result counts only against the planted truth; it never confirms the model's own conclusion.

The score is computed against the planted truth, not against the platform's own conclusion.

Precision, recall and F1 are recomputed from each run's log.

What constitutes a finding

Healthy network

✕ Error

Any finding reported here is an error.

Vulnerable network

✓ Finding

Recomputed from the log before it counts.

The finding is recomputed from the log; the conclusion is never taken on trust.

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A control task before testing the model

V0 is a control task, simple enough to be solved without inference. Both approaches must pass it.

1
Script passes

2
Model passes

Both checks pass

Both pass: the hypotheses may be read.

Only a control that both approaches pass permits the hypotheses to be read.

Calibration passed on the lab network (3 of 3 runs per arm). The held-out instance remains outstanding.

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The objective restated as hypotheses

The register is this list: each hypothesis, with the test that decides it.

H1	**Connected devices**	a link absent from public inputs
H2	**Heterogeneous formats**	a representation no parser covers
H3	**Novel contracts**	rights stated in unseen documentation
H4	**Vendor semantics**	a field whose meaning is only documented
H5	**Ambiguous refusals**	resuming the audit after a refusal
H6	**Hostile content**	an instruction obeyed from observed content

H1–H5 quantify a possible gain. H6 quantifies a risk.

Next steps

1. Execute V0 on a held-out instance
2. Develop a parameterisable test harness
3. Build the scenario families for H2, H3 and H4
4. Deploy a reproducible platform

Support requested

- Access to real vendor artifacts
- An adversarial review of the register

Order follows conditions, not dates.

Support line: AsiaCCS 2027 submission, 11 December 2026 (AoE).

Reserve — declared coverage gaps

- Physical impact and functional safety
- Device fragility under probing
- Real fleet scale
- Proprietary industrial protocols that are not simulated
- Complete multi-step exploitation chains
- Abstention and calibration
- Generic IT scenarios, deliberately excluded

A benchmark that states its gaps is stronger than one that claims full coverage.